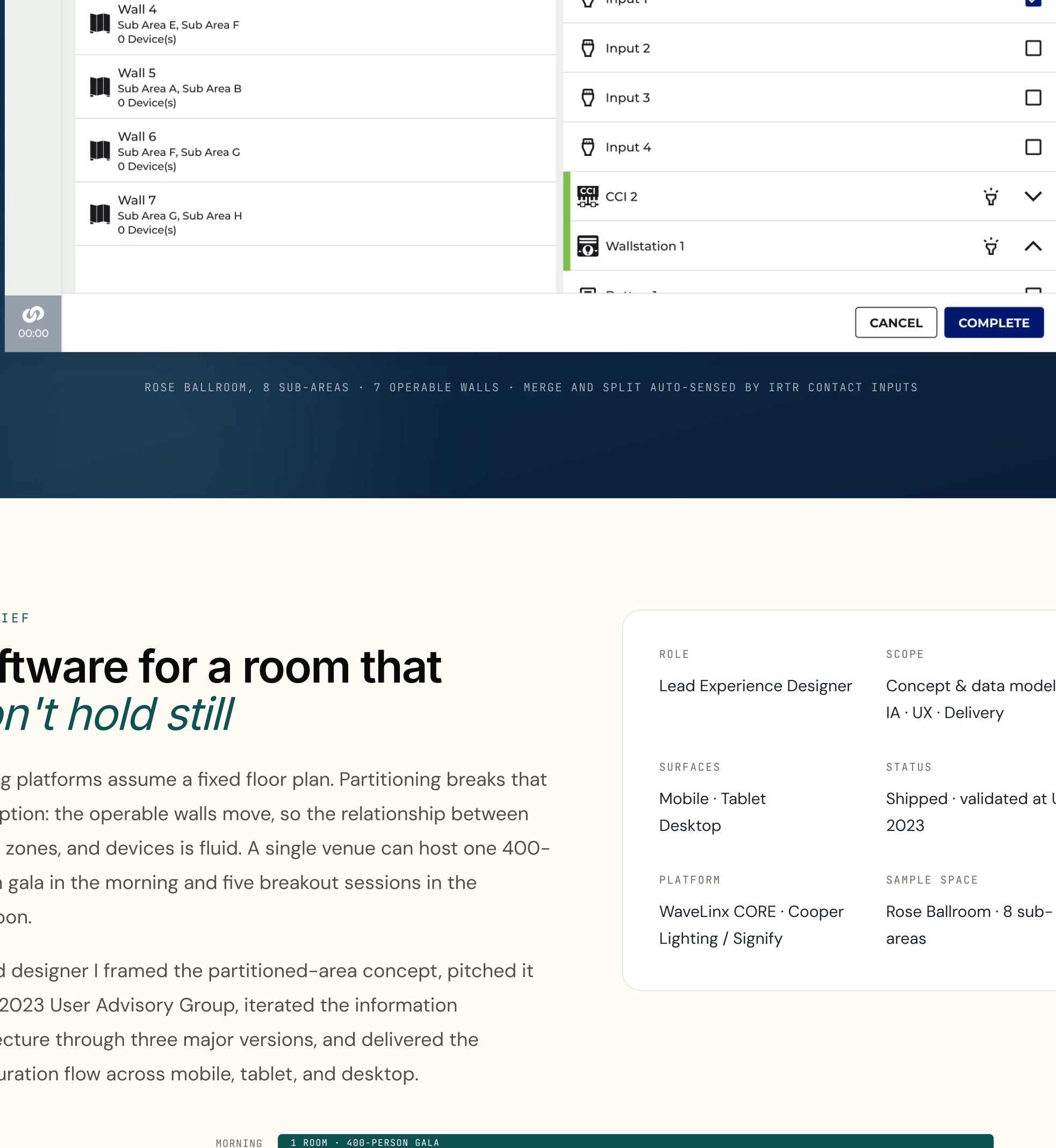


CONNECTED LIGHTING · COOPER LIGHTING / SIGNIFY

One room becomes many, the moment a *wall moves*

Dynamic Space Partitioning lets a single ballroom split into eight independently-lit rooms, and merge back, as operable walls open and close. I designed the system that models those moving walls, sub-areas, and devices into one configuration any installer can run.

[Read the full case study →](#)[See the flow](#)

THE BRIEF

Software for a room that *won't hold still*

Lighting platforms assume a fixed floor plan. Partitioning breaks that assumption: the operable walls move, so the relationship between rooms, zones, and devices is fluid. A single venue can host one 400-person gala in the morning and five breakout sessions in the afternoon.

As lead designer I framed the partitioned-area concept, pitched it at the 2023 User Advisory Group, iterated the information architecture through three major versions, and delivered the configuration flow across mobile, tablet, and desktop.

ROLE	SCOPE
Lead Experience Designer	Concept & data model IA · UX · Delivery
SURFACES	STATUS
Mobile · Tablet Desktop	Shipped · validated at UAG 2023
PLATFORM	SAMPLE SPACE
Wavelinx CORE · Cooper Lighting / Signify	Rose Ballroom · 8 sub- areas



ONE VENUE, ONE DAY, FROM THE BRIEF → 1 SALA → 5 BREAKOUTS · BLOCK LETTERS NOT TO TIME SCALE

CHALLENGE

A physical system, *made legible*

Three forces shaped every screen, each a constraint the system had to honor, not a feature to bolt on.

01 Walls aren't fixtures

A wall is a relationship between two rooms, not a device. No object in the platform modeled "these two sub-areas merge when this wall opens."

02 Merge & split, automatically

When a wall opens mid-event the lighting must unify instantly, and re-separate when it closes, without staff recalling scenes by hand.

03 Configure once, correctly

Sub-areas, walls, joins, devices and contact inputs are a lot to set up. One wrong mapping lights the wrong rooms, so the flow had to be guided and hard to get wrong.

"The wall moves. The lights should just know. I don't want to go find a laptop and open a program, I'm holding a phone."

COMMISSIONING AGENT · FIELD RESEARCH INTERVIEW

THE PAYOFF · LIVE FLOOR PLAN

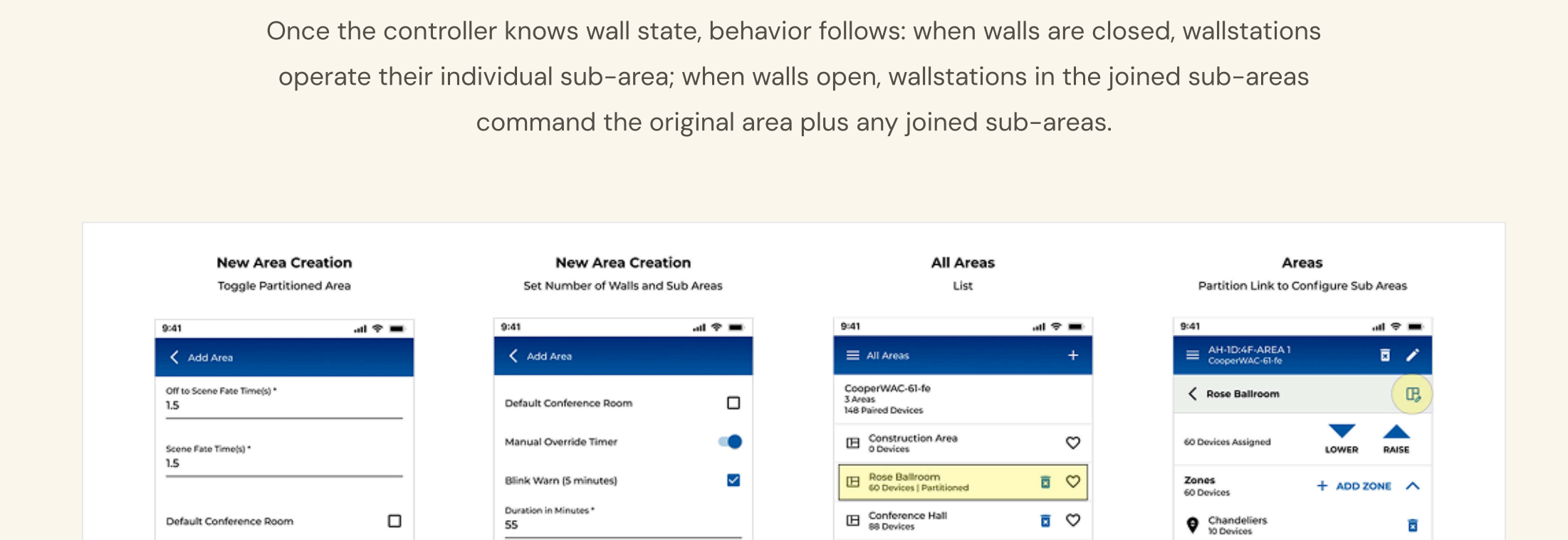
Open a wall. The rooms merge.

The Rose Ballroom, 8 sub-areas, 7 walls. Tap any wall to open or close it; sub-areas joined by an open wall fuse into a single lighting zone.

PRESETS · 8 Separate Rooms · 6 Ballrooms · 4 Ballrooms · 2 Events · 1 Grand Ballroom · 8 active lighting zones



THOUGHT PROCESS



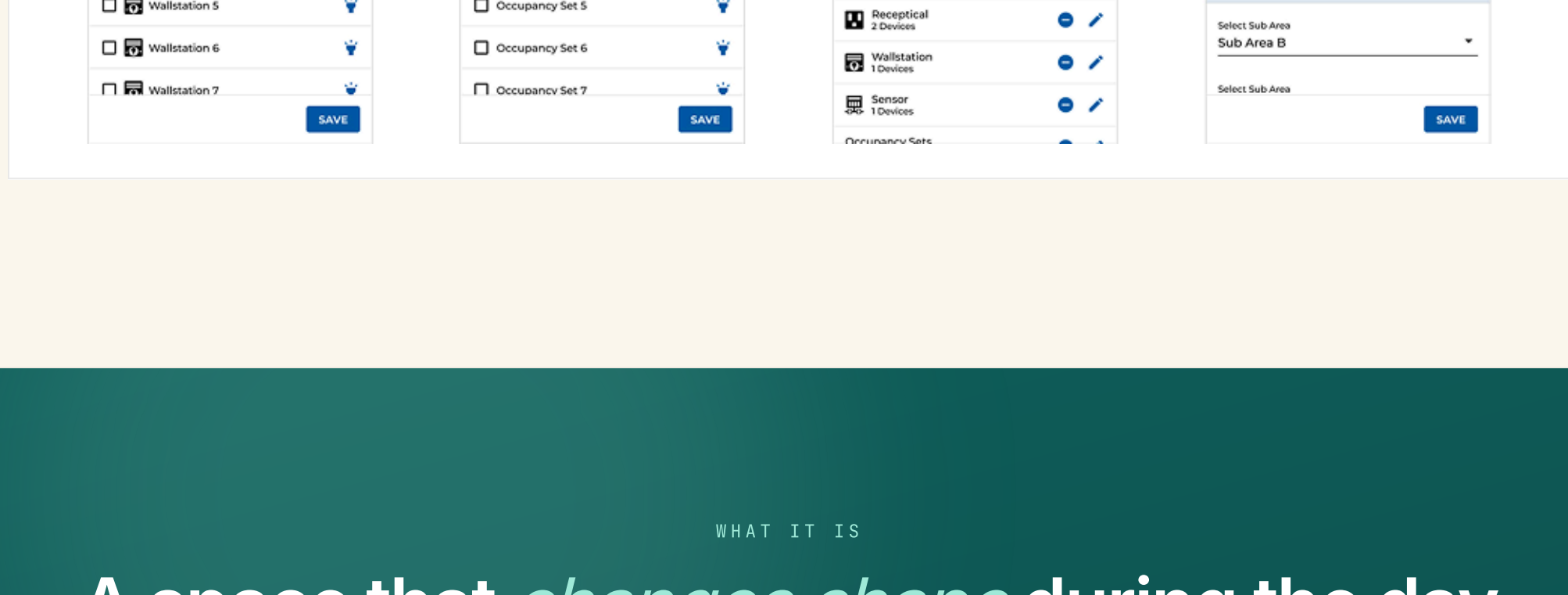
Thinking the room through before the app: every wall, sub-area, chandelier and cove seeped from the Rose Ballroom plan to the objects the configurator needs.

EVERY SURFACE

Commission on a phone, *operate on the wall*

An installer configures partitions on a phone at the panel; a venue manager runs the room from a wall-mounted wallstation. The same three-step model, sub-areas → walls → devices, scales to each screen.

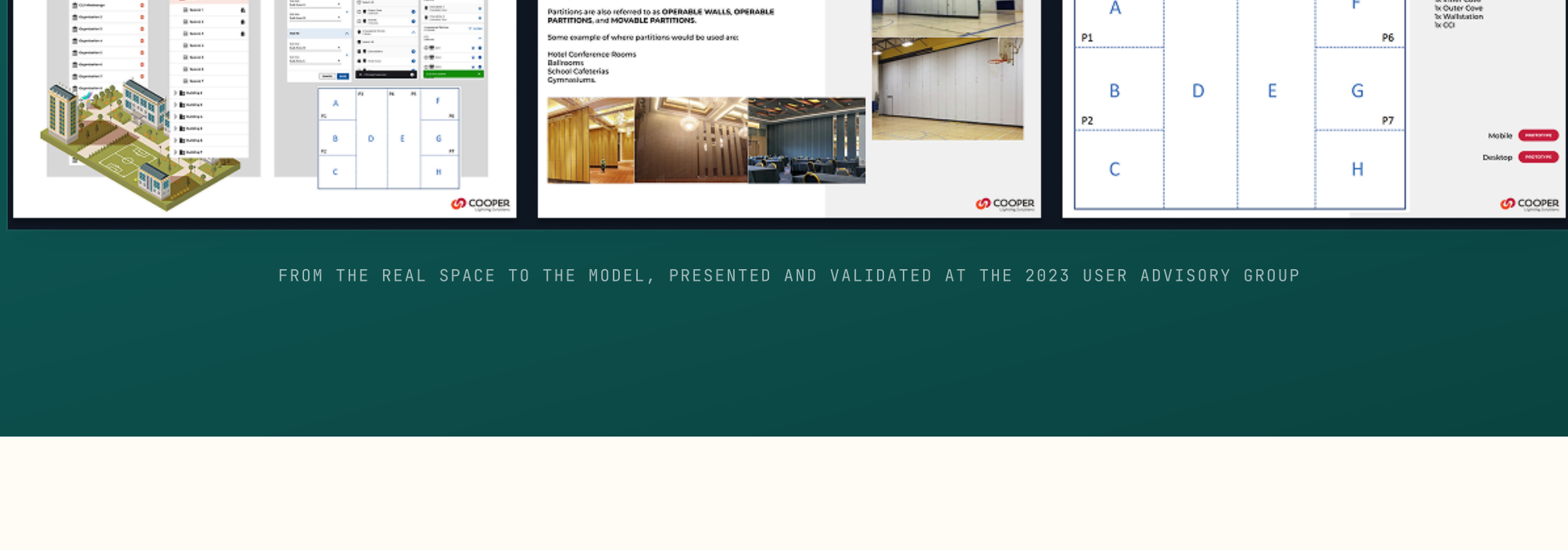
Once the controller knows wall state, behavior follows: when walls are closed, wallstations operate their individual sub-area; when walls open, wallstations in the joined sub-areas command the original area plus any joined sub-areas.



WHAT IT IS

A space that *changes shape* during the day

Partitioning divides large rooms into smaller areas with operable walls, so each room is lit on its own when closed and unified when open. It lives in hotel conference rooms, ballrooms, school cafeterias, and gymnasiums.



FROM THE REAL SPACE TO THE MODEL, PRESENTED AND VALIDATED AT THE 2023 USER ADVISORY GROUP

APPROACH

Rooms first, *walls as relationships*

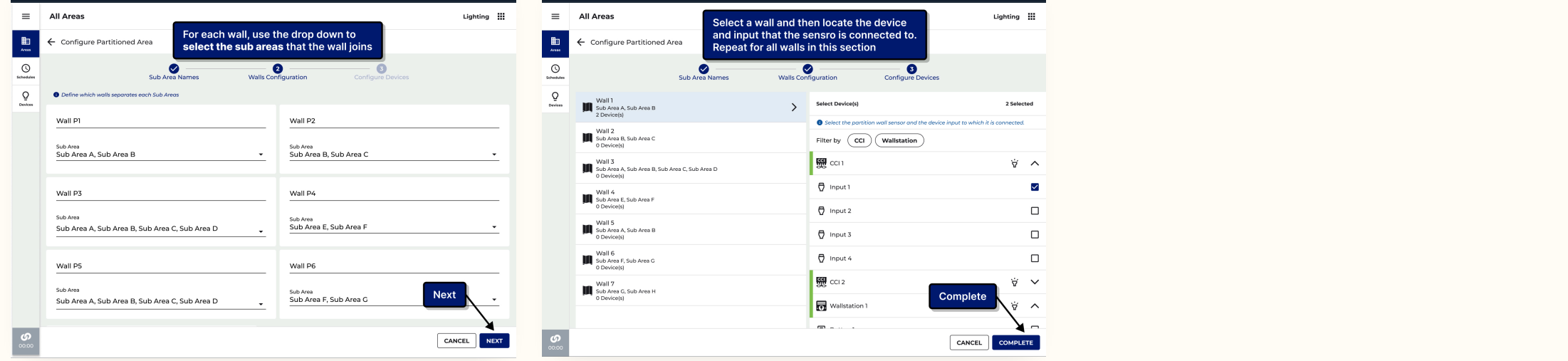
The design bet was to model the space the way the people running it think about it, not the way the hardware is wired. Get the mental model right and the contact-closure plumbing disappears behind a floor plan anyone can read.

- 01 Sub-areas as the primary object
Sub-areas own its own zones, devices, and occupancy; a venue manager describes the room this way, so the IA matches their mental model.
- 02 Walls demoted to a join
A wall is modeled as a relationship that says which sub-areas merge when it opens, not as another device to manage in a flat list.
- 03 Live state from contact inputs
Each wall is wired to a CCI / ITR contact input, so the Wall Status board reads each wall's real-time open or closed state, with manual override.
- 04 Guided wizard, runs itself
A 3-step wizard, sub-area names → wall configuration → devices, removes ambiguity at setup; after that the room re-lights itself as walls move.

THE CONFIGURATION FLOW

Five guided steps, *then it runs itself*

The shipped desktop wizard takes a room from the partitioning tool to fully wall-aware, defining sub-areas, the walls between them, and the contact input that senses each one.



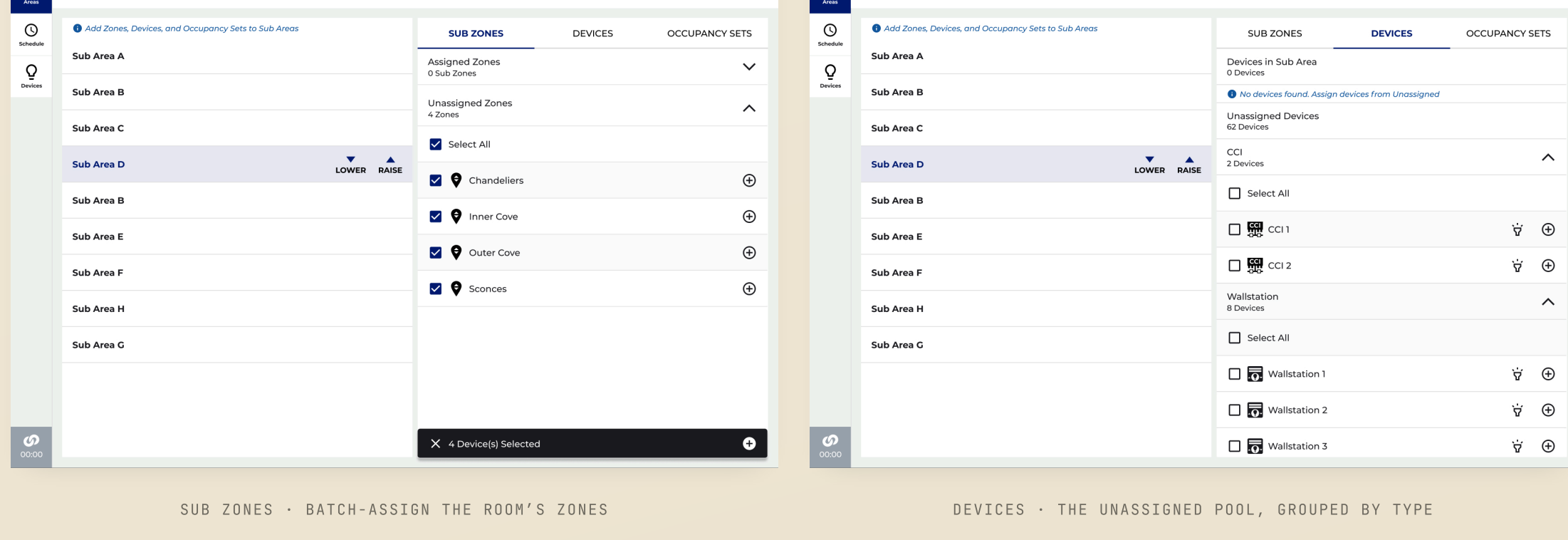
- 01 Create the Area
Tap + to add a new area, the entry point for any space being commissioned.
- 02 Define the Partitioned Area
Set the Partitioned type, then the wall and sub-area counts that describe the room.
- 03 Update Sub-Area Names
Name each room for the application so the floor plan reads in the venue's own language.

- 04 Define Each Wall's Joins
Pick the sub-areas each wall separates, the relationship that drives merge and split.
- 05 Assign the Wall Input
Map each wall to a CCI / ITR contact input so its open / closed state is sensed live.

SUB-AREA CONFIGURATION

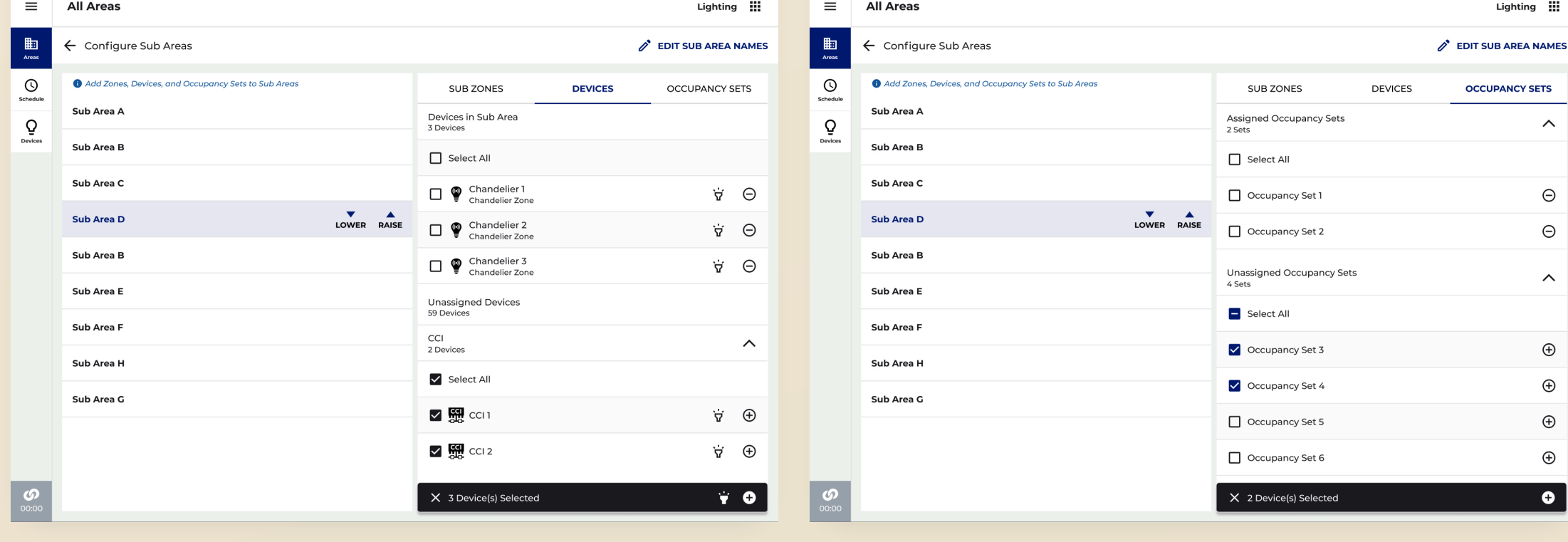
Zones, devices, occupancy, *one rail per room*

After the wizard, Configure Sub Areas fills each room: pick a sub-area on the left, and three tabs assign its lighting zones, devices, and occupancy sets from a shared unassigned pool, with batch select and an identify flash to confirm the physical fixture.



SUB ZONES · BATCH-ASSIGN THE ROOM'S ZONES

DEVICES · THE UNASSIGNED POOL, GROUPED BY TYPE

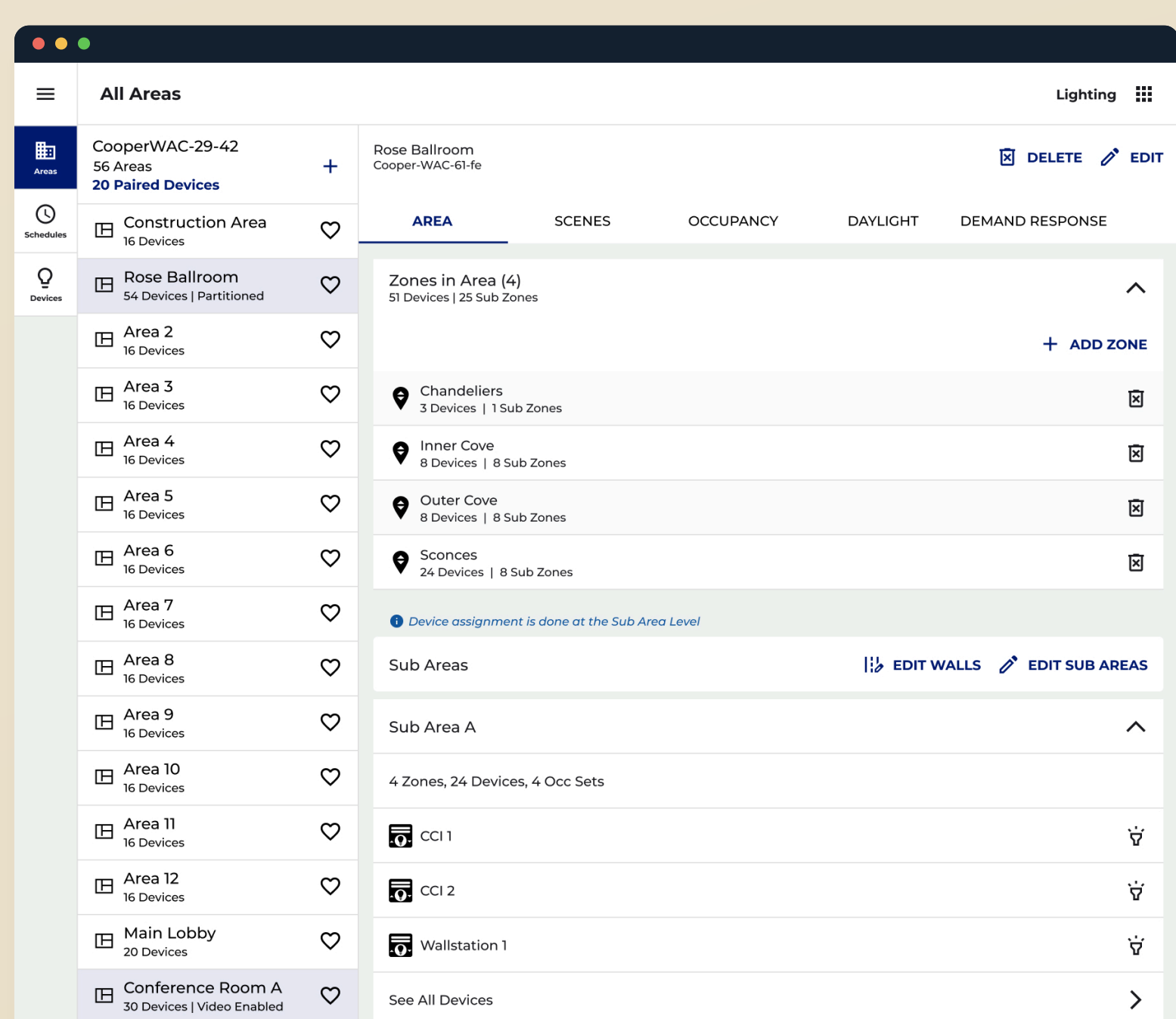


DEVICES · ASSIGNED, EACH WITH ITS ZONE

OCCUPANCY SETS · THE SAME GRAMMAR, PER ROOM

THE ASSEMBLED AREA

What configuration adds up to: one Rose Ballroom holding four zones, 54 devices, and eight sub-areas, with Sub Area A opened to its own share of 4 zones, 24 devices, and 4 occupancy sets. Walls and sub-areas stay one edit away.



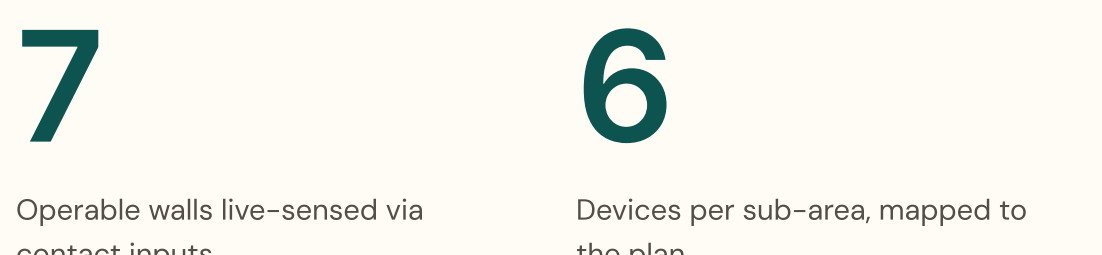
AREA · ROSE BALLROOM, PARTITIONED, EVERYTHING IN ITS PLACE

BY THE NUMBERS

A moving room, configurable once

- 8 Sub-areas configurable from one Rose Ballroom
- 7 Operable walls live-sensed via contact inputs
- 6 Devices per sub-area, mapped to the plan
- 3 Surfaces at parity: mobile, tablet, desktop

- 1st Mobile partitioning UX in connected commercial lighting
- 40–60% Reduction in on-site commissioning time, cited by field teams
- ~90% Fewer API calls on partition-area detail pages
- 11+ Critical usability issues resolved before build

BASELINE 1800
AFTER 600
REDUCTION IN TIME SPENT FROM THE STARTED 40-60%
METRICS · 40-60% PALE TO 40-60% REDUCTION IN TIME SPENT FROM THE STARTED 40-60%

WAVELINK CORE · COOPER LIGHTING / SIGNIFY · INDUSTRY-FIRST MOBILE PARTITIONING UX, VALIDATED AT UAG 2023 · METRICS FROM UAG SESSION DATA, ENGINEERING DOCS & FIELD INTERVIEWS.



The breakthrough wasn't a screen, it was deciding that rooms, not walls, should be the primary object, and letting the wall hardware fall behind a floor plan anyone can read.